A DETERMINANT BOUND OF 331 FOR THE TENTH DISCRIMINANT

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Abstract. We address the challenge of obtaining a small explicit upper bound for a hypothetical tenth imaginary quadratic discriminant of class number one, without using Heegner's methods or the known classification. We also make no quasi-Riemann-hypothesis assumption. We prove $D \leq 331$ for every negative fundamental discriminant $-D$ with class number one, improving the earlier bound $D \leq 498$. The earlier bound and elementary norm arguments leave six possible discriminants. To exclude them, we sharpen the rank criterion for an interpolation matrix by retaining the exact degree and vanishing-order bounds in its proof. This permits the determinant comparison to be applied with fewer evaluation points for each chosen polynomial space. A count of norm representations supplies the auxiliary inert primes. The arithmetic and spectral estimates are those used for the earlier bound; the six new numerical comparisons use directed rounding.

1. Introduction

Let $-D < 0$ be a fundamental discriminant, and write $\mathcal{O}_D$ for the ring of integers of $\mathbb{Q}(\sqrt{-D})$. Gauss's class number one problem asks when $\mathcal{O}_D$ has unique factorization, or equivalently when its ideal class number $h(-D)$ is one. Heegner [4], Baker [1] and Stark [5] proved that this occurs exactly for

$$D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Here we obtain a weaker bound by the determinant method, without using Heegner's methods or the known classification.

Theorem 1.1. *Let $-D < 0$ be a fundamental discriminant with $h(-D) = 1$. Then $D \leq 331$.*

The question is how small an explicit bound for a hypothetical tenth class-number-one discriminant can be obtained from such an argument. This is the setting of the challenge [7], which uses the class number one problem to test recent determinant arguments concerning Landau–Siegel zeros [3, 6]. Our earlier paper established $D \leq 498$ by comparing an upper bound for the norm of an interpolation determinant with the prime powers forced to divide that norm [2]. The present paper keeps that determinant and its spectral size estimate. We sharpen the rank criterion by retaining the exact degree and vanishing estimates in the earlier proof. For a fixed polynomial space and fixed derivative orders, this permits fewer evaluation points and improves the upper bound for the determinant.

1.1. **Overview of the proof.** Suppose that $h(-D) = 1$ and $D > 331$. Every element of $\mathcal{O}_D$ has the form $(a + b\sqrt{-D})/2$ with $a, b \in \mathbb{Z}$, so every non-rational element has norm at least $D/4$. If a prime p splits or ramifies, an ideal of norm p must be principal. Its generator cannot be a rational

Date: Working draft, 28 October 2026.
Key words and phrases. Imaginary quadratic fields, class number one, interpolation determinants, Frobenius, explicit bounds.

This work is based on an externally verified finding made by autonomous rStation agents (https://rstation.ai/). See the section on AI use for details.

integer, whose norm would be a square. Thus every prime $p < D/4$ is inert. In particular, D is odd, since otherwise 2 ramifies. Nor can D be composite: its least prime divisor would ramify and be at most $\sqrt{D} < D/4$. Since $-D$ is fundamental, the resulting odd prime D is congruent to 3 modulo 4; it cannot be 7 modulo 8, since then 2 splits. Hence D is prime with $D \equiv 3 \pmod 8$.

The bound $D \leq 498$ from [2] now leaves only

$$D \in \{347, 379, 419, 443, 467, 491\}.$$

We exclude these six values using the determinant estimates of [2] and the sharper rank criterion proved below.

We form the matrix by evaluating monomials $z^r w^s$ with $r + Ks < R$, and their normalized z-derivatives, at pairs $(z, \bar{z})$ with $z \in \mathcal{O}_D$. The parameter K determines the shape of this triangular set of exponents. At an inert prime p, conjugation agrees with Frobenius, so $z^p - \bar{z}$ is divisible by p. Expressing the monomials in terms of powers of $z^p - w$ therefore gives prime powers dividing every maximal minor, with a loss depending on the selected derivative orders. If one of these minors is nonzero, its norm is a positive integer with this prescribed divisibility. On the other hand, choosing the evaluation points near the origin gives an Archimedean upper bound for the same norm. We obtain a contradiction when the forced divisor exceeds that upper bound. The arithmetic estimate also uses larger inert primes: class number one lets us bound the loss from non-inert primes by counting norm values. The upper bound keeps the derivative columns at each node together, using the spectral estimates of [2] to account for their correlations.

The rank criterion determines how few points can be used while ensuring that a nonzero maximal minor exists. If the polynomial space has dimension M, and we take J derivative orders at each of n points, the necessary dimension inequality is $nJ \geq M$. The sufficient inequality stated in [2] does not hold uniformly over the auxiliary-prime intervals used here. Its proof, however, contains more precise information.

As in that proof, we rescale the polynomial rows and derivative columns by known powers of an auxiliary inert prime before reducing modulo the prime. A relation among the reduced rows would give a polynomial in two variables with prescribed local vanishing; its second variable records the powers of $z^p - w$. Applying a differential operator and taking a determinant of coefficients then produces a nonzero polynomial in one variable. We bound its degree and the multiplicity of its zeros, retaining their dependence on the degree of the original relation in its second variable. These more precise bounds show that the total multiplicity exceeds the degree, even when the earlier sufficient inequality does not apply. This proves full row rank for the smaller node counts.

We still need an auxiliary inert prime for which this rank criterion applies. The required primes are larger than $D/4$, where the elementary argument no longer guarantees inertness. Our chosen intervals lie above D, so their primes cannot ramify. If a prime in one of these intervals were split, class number one would represent it by the norm form

$$p = x^2 + xy + \frac{D+1}{4} y^2.$$

Counting the possible representations bounds the number of split primes in the interval. When y is even, the norm is congruent to x^2 modulo 2; an odd prime norm therefore requires x odd, so the admissible values of x are spaced by two. We use this parity refinement for the two intervals needed when $K = 17$. Since the count is smaller than the number of primes in the interval, at least one prime there is inert. We also place each interval above twice the radius of the disk containing the nodes. Distinct nodes then remain distinct after reduction at every prime in the interval, as the rank criterion requires.

For $D = 491, 467, 443, 419$ we use $K = 43$. For $D = 379$ and $D = 347$, the stronger rank criterion permits $K = 17$. At fixed R, reducing K allows more powers of the conjugate variable, changing both the divisibility and the Gram-matrix estimate. Their combined comparison is more

favorable at these two discriminants. For each chosen polynomial space, the sharper criterion brings the number of points close to the minimum M/J. With these choices, the arithmetic lower bound exceeds the Meixner–Schur upper bound for each of the six candidates. All six values are therefore excluded, proving Theorem 1.1.

The rest of the paper is organized around these steps. Section 2 recalls the determinant inequality and states the imported size estimate. Section 3 proves the exact rank criterion. Section 4 supplies the auxiliary inert primes and fixes the three parameter families, while Section 5 gives the six numerical comparisons. The accompanying calculations and source code are available at https://github.com/dualverse-ai/tenth_discriminant.

2. The determinant comparison

We recall the determinant and the estimates from [2] that will be used. The comparison applies to every nonzero maximal minor. Thus the rank argument in the next section need only establish that such a minor exists; it need not identify its columns.

2.1. **The matrix and its normalization.** Assume that $h(-D) = 1$ and $D > 331$. Write $\omega = (1+\sqrt{-D})/2$ and $\mathcal{O}_D = \mathbb{Z}[\omega]$. Choose integers $K \geq 2$, $S \geq 1$ and $1 \leq J \leq R := KS$, and put

$$\Lambda = \{(r,s) \in \mathbb{Z}^2_{\geq 0} : r + Ks < R\}, \qquad M = |\Lambda| = \frac{R(S+1)}{2}. \tag{2.1}$$

Let $z_1, \dots, z_n$ be the n elements of $\mathcal{O}_D$ nearest to the origin, with ties broken arbitrarily. We use the Hasse derivatives in z, with the second variable w held fixed: $\partial_z^{[j]} P$ is the coefficient of t^j in $P(z+t, w)$. The matrix is

$$V_{(r,s),(i,j)} = \binom{r}{j} z_i^{r-j} \bar{z}_i^{\,s}, \qquad (r,s) \in \Lambda, \quad 1 \leq i \leq n, \quad 0 \leq j < J. \tag{2.2}$$

The entry is zero if $j > r$. Dividing ordinary derivatives by $j!$ removes their factorial factors while retaining algebraic-integer entries. For any selection C of M columns, let $\Delta_C = \det V_C$.

The nearest-node estimate of [2, Lemma 2.5] gives

$$|z_i| \leq \rho_i := \sqrt{\frac{i\sqrt{D}}{2\pi}} + \delta, \qquad \delta = \frac{\sqrt{D+4}}{4}. \tag{2.3}$$

The lattice has covolume $\sqrt{D}/2$ and covering radius at most δ: its horizontal rows have vertical spacing $\sqrt{D}/2$, and the points on each row have spacing one. The disk of radius $\rho - \delta$ is therefore covered by Voronoi cells of lattice points in the disk of radius ρ. Comparing areas shows that this latter disk contains at least $2\pi(\rho-\delta)^2/\sqrt{D}$ lattice points, proving (2.3).

Put $\rho = \rho_n$, $\theta = M/(nJ)$, and

$$W = \frac{R^2 S}{3}, \qquad W_1 = 2\sum_{\Lambda} r, \qquad W_2 = 2\sum_{\Lambda} s. \tag{2.4}$$

The ratio θ is the proportion of available columns used by a maximal minor. In particular, $\theta \leq 1$ is necessary for full row rank. Direct summation over the triangle gives

$$\frac{W_1 + W_2}{W} = 1 + \frac{1}{K} + \frac{3(K-1)}{2KS} + \frac{K-5}{2KS^2}. \tag{2.5}$$

2.2. **The arithmetic estimate.** We use the local divisibility formulas of the earlier paper, with all quantities evaluated at the present parameters. Their general derivation is given in Appendices A and B of [2]. We record the hypotheses and the meaning of the resulting terms, since the finite comparisons in that paper use different parameters.

Suppose throughout this subsection that $D > 163$ is prime, $D \equiv 3 \pmod 8$, and $h(-D) = 1$. Put $m_D = (D+1)/4$. Let P be the prime cutoff in the arithmetic sums. The parameters used here satisfy

$$\begin{aligned} 2 \le K < D/4, \quad R = KS, \quad 1 \le J \le R, \quad M \le nJ, \\ P \ge \max(S, J-1), \qquad F_{\max} := \lfloor P/m_D \rfloor < m_D^4. \end{aligned} \tag{2.6}$$

The local divisibility estimates give a number $\mathcal{A}$ and a derivative loss $b(j)$ such that every nonzero maximal minor satisfies

$$W\mathcal{A} - 2\sum_{C} b(j) \le \log |\det V_C|^2. \tag{2.7}$$

We recall their construction to specify how the estimates apply at the present parameters. An element is *primitive* if it is not divisible in $\mathcal{O}_D$ by a rational integer greater than one.

All primes below K are then inert. The small-prime estimates apply without exception, and the cutoff P includes the derivative debits, the ramified prime, and the primes contributing to the conjugate-variable estimates. Summing the local row bounds gives

$$\mathcal{A} = \mathcal{A}_0 - L_{\mathrm{norm}} + C_2 + C_3. \tag{2.8}$$

Here $\mathcal{A}_0$ provisionally treats primes up to P as inert. Class number one represents every non-inert prime as the norm of an algebraic integer. Summing over primitive norm representatives therefore gives an upper bound L_{norm} for the necessary correction, retaining the split and ramified local contributions. This sum also includes composite norm values. The nonnegative terms C_2, C_3 correct this overestimate for products of two or three prime factors, with the representation multiplicities supplied by unique factorization. Omitting other composite corrections only weakens the lower bound.

Some split-prime column contributions are retained individually. Let $\mathcal{P}_s(D)$ be the split primes at most q_{cap}, where the cutoff satisfies

$$q_{\mathrm{cap}} \le \min(J-1, F_{\max} m_D, m_D^4 - 1). \tag{2.9}$$

Define the remaining derivative loss and its factorial remainder by

$$\begin{aligned} b(j) &= \sum_{q \le j} \lfloor j/q \rfloor \log q - \sum_{q \in \mathcal{P}_s(D)} \lfloor j/q \rfloor \log q, \\ F(j) &= \log(j!) - b(j) = \sum_{q} \sum_{a \ge 2} \lfloor j/q^a \rfloor \log q + \sum_{q \in \mathcal{P}_s(D)} \lfloor j/q \rfloor \log q. \end{aligned} \tag{2.10}$$

Sums indexed by q are over primes. The split-prime contributions outside this window are bounded uniformly over column selections and already included in (2.8).

The prime sums in the window are bounded using norm representatives and composite products. Every prime factor of a primitive norm is at least m_D. Thus a composite primitive norm in (2.9) has at most three prime factors, counted with multiplicity, and each factor is at most $F_{\max}$. To obtain the factor list up to $F_{\max}$, the finite enumeration first removes reducible primitive elements. A reducible primitive element of norm at most $F_{\max} < m_D^4$ has an irreducible factor of norm at most $\sqrt{F_{\max}} < m_D^2$, and a remaining factor of norm at most $F_{\max}/m_D$. Appendix B of [2] proves that these ranges construct the complete factor list. Consequently the subtraction includes every composite in the window, even for $P > m_D^4$.

Using Appendix B of [2], we choose a nonnegative lower bound σ for

$$\sum_{\substack{q,\ a\ge 2\\ q^a\le J-1}} \frac{\log q}{q^a} + \sum_{q\in\mathcal{P}_s(D)} \frac{\log q}{q},$$

and upper bounds h_j for $H(j) = j\sigma - F(j)$. These bounds determine c by the capacity calculation below. The explicit finite weights, slope bounds and remainder estimates are those in Appendix B of [2]. The hypotheses above hold for all three parameter families below.

2.3. **Combining the derivative terms.** Legendre's formula writes $\log(j!)$ as $\sum_q \sum_{a\ge1} \lfloor j/q^a \rfloor \log q$. Its first-power terms cancel the corresponding derivative loss in $b(j)$, leaving the higher prime powers and split-prime contributions in $F(j)$. We use the remainder $H(j) = j\sigma - F(j)$ defined above, and choose a constant c satisfying

$$2\sum_{(i,j)\in C} H(j) \le Mc \qquad (|C| = M). \tag{2.11}$$

This bound uses the fact that each derivative order is available at only n nodes. Specifically, if $h_{(1)} \ge \cdots \ge h_{(J)}$ are upper bounds for $H(0), \dots, H(J-1)$ arranged in decreasing order, and $M = nt + r$ with $0 \le r < n$, one may take

$$Mc = 2n\sum_{a=1}^{t} h_{(a)} + 2rh_{(t+1)}.$$

The last term is omitted when $r = 0$. The formulas in Appendix B of [2] construct σ and these upper bounds from prime powers and norm values, so the calculation does not require identifying the split primes at the candidate discriminant.

Set $u_i = z_i/\rho$ and $\ell = \log(R/\rho) - \sigma$. Define the normalized matrix

$$B_{(r,s),(i,j)}(\ell) = \frac{(r)_{\underline{j}}}{R^j} e^{\ell j} u_i^{r-j} \bar{u}_i^{\,s}, \tag{2.12}$$

where $(r)_{\underline{j}} = r(r-1)\cdots(r-j+1)$. Exact diagonal scaling gives

$$B_C(\ell) = \operatorname{diag}_\Lambda(\rho^{-r-s})\, V_C\, \operatorname{diag}_C(j!e^{-\sigma j}).$$

Taking squared absolute determinants and using (2.7) and (2.11), we obtain

$$W\mathcal{A} \le (W_1 + W_2)\log\rho + \log|\det B_C(\ell)|^2 + Mc. \tag{2.13}$$

Since $e^{\ell j}$ scales column (i,j) and $j \ge 0$, replacing ℓ by an upper bound only increases $|\det B_C(\ell)|^2$. It therefore suffices to bound the normalized determinant uniformly over C.

2.4. **The imported size estimate.** We use the Meixner and Schur estimates from [2, Section 4]. Their role is to retain the correlations between derivative columns at one node. We recall the quantities entering the bound so that its application at the new parameters is explicit.

Choose $a > 0$, $0 < v < 1$, $\beta < 0$, and put $b = aR$. Weight row (r,s) by $e^{-\lambda_{r,s}/2}$, where

$$e^{-\lambda_{r,s}} = \frac{(b)_r}{r!} v^r e^{-\beta s}, \qquad (b)_r = b(b+1)\cdots(b+r-1).$$

Let $Y = \operatorname{diag}(e^{-\lambda_{r,s}/2})B(\ell)$, and let $G_i = Y_i^* Y_i$ be the Gram matrix of its J columns at node i. Fischer's inequality gives

$$\log|\det B_C(\ell)|^2 \le \sum_\Lambda \lambda_{r,s} + \sum_i \log\det G_i[C_i, C_i], \tag{2.14}$$

where C_i is the set of orders selected at that node. By [2, Lemma 4.1], each principal minor of G_i is a polynomial with nonnegative coefficients in $q = |u_i|^2$; therefore it increases when the node is moved radially outward.

Write $x_+ = \max(0, x)$. At an upper squared radius $q > 0$, set

$$d_q = (\log q - \beta)_+, \qquad v_q = ve^{-d_q/K}, \qquad w = v_q q, \qquad \mathcal{B}_q = Se^{Sd_q}.$$

After a diagonal unitary conjugation removing the node angle, the Gram matrix is bounded in the positive semidefinite order by

$$\mathcal{G}_{jk}(q) = \mathcal{B}_q \left(\frac{e^{\ell}}{R\sqrt{q}} \right)^{j+k} \sum_{r \geq 0} \frac{(b)_r}{r!} w^r (r)_{\underline{j}} (r)_{\underline{k}}, \qquad 0 \leq j, k < J. \tag{2.15}$$

The conjugation preserves every principal minor. Here $0 < w < 1$ for the parameters below. The negative-binomial moments in this formula have an explicit factorization in Meixner polynomials. This supplies two bounds for an arbitrary principal minor: one uses the largest singular values of an affine composition operator, and the other its complementary singular product. A third bound sums all principal minors of the prescribed size, using the Schur-function Cauchy identity. These are respectively equations (4.10), (4.11), and (4.12) of [2]. Each holds for every subset of derivative orders; no particular subset is assumed to maximize the determinant.

Let $F_q(m)$ be the minimum of these three upper bounds for the logarithm of an m-column principal minor, allowing the finite parameter choices in Appendix C of [2]. For a chosen $a_0 > 0$, take $\Phi(q) \geq 0$ such that

$$F_q(m) \leq a_0 Rm + R^2 \Phi(q) \qquad (0 \leq m \leq J). \tag{2.16}$$

We divide the range of m/R into N_m equal intervals and establish the bound on each whole interval before minimizing over estimates. For the singular-value bounds, the sum of the m largest column prices is bounded using N_z equal intervals of $j/R \in [0, \varepsilon]$, where $\varepsilon = J/R$. Appendix C of [2] bounds the price and the number of orders in each interval. Summing (2.16) in (2.14) uses $\sum_i |C_i| = M$, rather than allowing every node to choose its worst number of columns independently.

To bound the radii, put

$$\eta = \frac{\delta}{\sqrt{n\sqrt{D}/(2\pi)}}, \qquad Q(t) = \left(\frac{\sqrt{t} + \eta}{1 + \eta} \right)^2, \qquad \varepsilon = J/R.$$

Then $|u_i|^2 \leq Q(i/n)$. Divide $(0, 1]$ into N_t equal intervals and move the nodes in the kth interval to squared radius $Q(k/N_t)$. Each interval contains at most $n/N_t + 1$ nodes. The resulting bound is

$$U_{\mathrm{var}} = U_{\mathrm{row}} + \frac{3a_0(S+1)}{2S} + 3\left(\frac{1 + 1/S}{2\theta\varepsilon N_t} + \frac{1}{S} \right) \sum_{k=1}^{N_t} \Phi(Q(k/N_t)), \tag{2.17}$$

where U_{row} is an upper bound for $W^{-1} \sum_\Lambda \lambda_{r,s}$. In particular, $W^{-1} \log |\det B_C(\ell)|^2 \leq U_{\mathrm{var}}$ for every C. For the finite formulas in Appendix C of [2], we require $R > 3$, $aR > 1$, $0 < a \leq 1$, $0 < \varepsilon \leq 1$, $\varepsilon(a + 2\varepsilon) \leq 1$, and

$$-(a+1)\log(a+1) + a \log a - \log v < 0.$$

The last condition gives the signed row-sum bound. For the Schur bound at radius q, put

$$\zeta = \ell + \tfrac{1}{2} \log v_q - \log(1 - w), \qquad z_* = a^{-2} e^{-2\zeta}.$$

Its finite evaluation is used when $z_* < 1$ and $\varepsilon(a + \varepsilon) z_* < a^2 (1 - z_*)^2$. These are respectively the moment and concavity conditions in Appendix C of [2]. The other two bounds remain available wherever the latter conditions do not hold.

Proposition 2.1 (Determinant comparison)**.** *Assume the arithmetic estimates* (2.7) *and* (2.11)*, and the size estimate* (2.17)*. If V has full row rank, then*

$$\mathcal{A} \le \mathcal{U} := \frac{W_1 + W_2}{W} \log \rho + U_{\mathrm{var}} + \frac{Mc + 2M \log M}{W}. \tag{2.18}$$

Proof. Full row rank supplies a nonzero minor. Apply (2.13) and the uniform size bound to that minor. The nonnegative term $2M \log M/W$ retains the earlier numerical normalization [2]. □

Thus a positive lower bound for $G := \mathcal{A} - \mathcal{U}$ excludes the proposed field. The arithmetic and size estimates depend on the actual node count n throughout. In particular, reducing n changes both the disk radius and the capacity bound for the derivative orders.

3. An exact rank criterion

The determinant comparison requires the matrix V to have full row rank. The necessary dimension inequality $nJ \ge M$ does not by itself establish this. We use the same reduction at an inert prime as in the earlier bound, but retain the individual degree and multiplicity estimates throughout the argument. This gives a stronger sufficient criterion at the parameters needed here.

The obstruction to direct reduction is the identity $\bar{z} = z^p$ modulo an inert prime p. Thus $z^p - w$ vanishes at every reduced node, and some of the rows become dependent. Dividing out known powers of p before reducing preserves more information. A relation among the resulting rows would produce a polynomial with prescribed vanishing along several curves. We will construct from it a nonzero polynomial in one variable, and show that its zeros have total multiplicity greater than its degree.

Theorem 3.1 (Exact rank criterion)**.** *Let $K \ge 2$ and $S \ge 1$ be integers, put $R = KS$, and let $1 \le J \le R$. Suppose that p is inert in $\mathcal{O}_D$, that*

$$K < p < R \le p^2,$$

and that the nodes $z_1, \dots, z_n \in \mathcal{O}_D$ have distinct residues modulo $p\mathcal{O}_D$. Set $d = \lfloor (R-1)/p \rfloor$, and define

$$\begin{aligned} D_m &= p \left\lfloor \frac{R - mp - 1}{K} \right\rfloor + ((R - mp - 1) \bmod K) \qquad (0 \le m \le d), \\ B_e &= \sum_{m=0}^{e} (D_m - m) \qquad (0 \le e \le d). \end{aligned} \tag{3.1}$$

For $l \ge 0$ put $L_l = \min(p, \max(0, J - lp))$, and let

$$\mu_e = \sum_{l=0}^{e} \max(0, L_l - e + l) \qquad (0 \le e \le d). \tag{3.2}$$

If

$$n\mu_e > B_e \qquad (0 \le e \le d), \tag{3.3}$$

then the matrix (2.2) *formed with these nodes has full row rank.*

Here D_m is the largest degree of a coefficient polynomial arising from layer m of the reduced relation. The quantity B_e bounds the degree of the one-variable polynomial constructed from a relation of degree e in the second variable, while μ_e bounds its order of vanishing at each node. Keeping these quantities separately for every e avoids the losses in the earlier sufficient criterion.

3.1. **A relation after reduction modulo an inert prime.** Put $g = z^p - w$. The polynomials

$$g^m z^a w^s, \qquad 0 \le m \le d, \quad 0 \le a < p, \quad a + Ks < R - mp, \tag{3.4}$$

form an integral basis of the space spanned by the triangular support $z^r w^s$, $r + Ks < R$. Indeed, writing $r = mp + a$ identifies their leading monomials with the original basis. Every other term replaces a factor z^p by $-w$, decreasing the weighted degree by $p - K > 0$. The change of basis is therefore unitriangular.

At node i write

$$g(z_i + t, \bar{z}_i) = pU_i(t) + t^p, \qquad U_i(t) = \frac{(z_i + t)^p - t^p - \bar{z}_i}{p} \in \mathcal{O}_D[t].$$

The constant coefficient is integral because p is inert, and the remaining coefficients are integral because p divides $\binom{p}{v}$ for $0 < v < p$. Divide each row in layer m by p^m, and multiply each column of derivative order $j = lp + v$, $0 \le v < p$, by p^l. Denote the resulting matrix by $\widetilde{V}$. Its entries are still integral: a term using t^{pq} in the expansion of $(pU_i + t^p)^m$ can contribute to order j only if $q \le l$, and its coefficient after rescaling contains p^{l-q}. Only $q = l$ can survive reduction modulo p.

Write $k = \mathcal{O}_D/p\mathcal{O}_D$, which is a field, and put $\xi_i = z_i \bmod p$ and $u_i(z) = (U_i \bmod p)(z - \xi_i)$. For $l \le m$ the reduced entry is

$$\binom{m}{l}[t^v](u_i(\xi_i + t)^{m-l}(\xi_i + t)^a \xi_i^{ps}); \tag{3.5}$$

for $l > m$ it is zero. The basis change and the scalings are invertible in characteristic zero. Consequently, full row rank of $\widetilde{V} \bmod p$ will imply full row rank of V. These scalings are used only for this purpose; the size and divisibility estimates concern the original matrix.

Suppose that the reduced rows satisfy a nontrivial relation, with coefficients $c_{m,a,s} \in k$. Define

$$F(z, u) = \sum_{m=0}^{d} H_m(z) u^m, \qquad H_m(z) = \sum_{a,s} c_{m,a,s} z^{a+ps}. \tag{3.6}$$

The exponents $a + ps$ within a layer are distinct because $0 \le a < p$. Thus $F \ne 0$. Let $e = \deg_u F \le d < p$, where the last inequality follows from $R \le p^2$. For $t = R - mp - 1$, the maximum of $a + ps$ subject to $a + Ks \le t$ and $0 \le a < p$ is attained at $s = \lfloor t/K \rfloor$, $a = t \bmod K$, since $p > K$. It follows that

$$\deg H_m \le D_m. \tag{3.7}$$

We will also use $D_m - D_{m+1} \ge p$ for $m < d$. To see this without manipulating the floors, take a monomial attaining D_{m+1} and increase s by one. Its weighted degree increases by $K < p$, so it is admissible in layer m, and its degree $a + ps$ increases by p.

There are L_l derivative orders $j = lp + v$ with $0 \le j < J$. Since $(\xi_i + t)^{ps} \equiv \xi_i^{ps} \pmod{t^p}$ in characteristic p, we may replace ξ_i^{ps} in (3.5) by $(\xi_i + t)^{ps}$ when extracting coefficients of order less than L_l. The row relation therefore gives

$$\operatorname{ord}_{z=\xi_i}((\partial_u^{[l]} F)(z, u_i(z))) \ge L_l \qquad (1 \le i \le n,\ 0 \le l \le e). \tag{3.8}$$

Here $\partial_u^{[l]}$ is the Hasse derivative, defined by $\partial_u^{[l]} u^m = \binom{m}{l} u^{m-l}$.

3.2. **A nonzero polynomial of controlled degree.** Differentiating the identity defining U_i before reducing modulo p gives

$$u_i'(z) = z^{p-1} - (z - \xi_i)^{p-1}. \tag{3.9}$$

Thus differentiation along the different curves $u = u_i(z)$ agrees, up to a term vanishing to order $p - 1$ at the relevant node, with the common operator

$$\mathcal{D} = \partial_z + z^{p-1}\partial_u.$$

On $k(z)[u]$ this operator satisfies

$$\mathcal{D}^p = -\partial_u. \tag{3.10}$$

Indeed, the pth iterate of a derivation in characteristic p is a derivation. It sends z to zero and u to $(p-1)! = -1$, which determines its action on $k(z)[u]$.

We claim that $F, \mathcal{D}F, \dots, \mathcal{D}^e F$ are linearly independent over $k(z)$. Otherwise, since $F \neq 0$, let $1 \leq r \leq e$ be the first index at which a dependence occurs. The span of $F, \dots, \mathcal{D}^{r-1}F$ is then stable under $\mathcal{D}$, including when $\mathcal{D}$ differentiates rational coefficients. By (3.10) it is also stable under ∂_u. It must therefore contain $F, \partial_u F, \dots, \partial_u^e F$. These are independent: since $e < p$, they have distinct degrees $e, e-1, \dots, 0$ in u, with nonzero leading coefficients. This contradicts the dimension $r \leq e$.

Consequently the coefficient determinant

$$W_F(z) = \det\big([u^m]\mathcal{D}^j F\big)_{0 \leq j,m \leq e} \tag{3.11}$$

is a nonzero polynomial. The identity

$$[u^m]\mathcal{D}G = ([u^m]G)' + (m+1)z^{p-1}[u^{m+1}]G$$

and $D_m - D_{m+1} \geq p$ give, by induction on j,

$$\deg\big([u^m]\mathcal{D}^j F\big) \leq D_m - j.$$

Terms with $m > e$ are zero, and a negative degree bound means the corresponding entry is zero. Each determinant term uses every row and every column exactly once, so

$$\deg W_F \leq \sum_{m=0}^{e} D_m - \sum_{j=0}^{e} j = B_e. \tag{3.12}$$

3.3. **The multiplicity at each node.** Fix i. Expressing each row polynomial of (3.11) in powers of $u - u_i(z)$ makes a unitriangular change of columns. It preserves the determinant and replaces its entries by

$$T_{j,l}^{(i)}(z) = (\partial_u^{[l]}\mathcal{D}^j F)(z, u_i(z)).$$

We first show that

$$\operatorname{ord}_{\xi_i} T_{j,l}^{(i)} \geq \max(0, L_l - j). \tag{3.13}$$

Let $\mathcal{D}_i = \partial_z + u_i'(z)\partial_u$. By (3.9), $\mathcal{D} - \mathcal{D}_i = (z-\xi_i)^{p-1}\partial_u$. Repeated application lowers the order of this error by at most one at each step; hence, for every polynomial G and $0 \leq j < p$,

$$\mathcal{D}^j G - \mathcal{D}_i^j G \in (z-\xi_i)^{p-j} k[z,u].$$

Both operators commute with the Hasse derivatives in u. After substituting $u = u_i(z)$, the entry $T_{j,l}^{(i)}$ therefore agrees, modulo $(z-\xi_i)^{p-j}$, with the jth ordinary derivative of $(\partial_u^{[l]}F)(z, u_i(z))$. Equation (3.8) bounds its order from below by $L_l - j$ when this is positive. Since $L_l \leq p$, the error has at least the same order, proving (3.13).

To use all these bounds together, consider a term in the determinant expansion. Its order is at least $\sum_{l=0}^{e}(L_l - j_l)_+$, where the j_l are a permutation of $0, \dots, e$. For $a \geq b$ and $x \geq y$ one has

$$(a-x)_+ + (b-y)_+ \leq (a-y)_+ + (b-x)_+.$$

Thus pairing the largest L_l with the largest row index minimizes this sum. Since $L_0 \geq \cdots \geq L_e$, its minimum is $\sum_{l=0}^{e}(L_l - e + l)_+ = \mu_e$. Every determinant term has at least this order, and so does their sum:

$$\operatorname{ord}_{\xi_i} W_F \geq \mu_e.$$

Since W_F is nonzero and the residues are distinct, (3.12) gives

$$n\mu_e \le \deg W_F \le B_e,$$

contrary to (3.3). This proves Theorem 3.1.

4. Auxiliary primes and the choice of parameters

The rank theorem requires an inert prime large enough to distinguish the nodes, but smaller than R and with $R \le p^2$. We obtain it by counting norm representations in a short interval. This uses the class number one hypothesis in the same way as the arithmetic estimate: every split prime has to be the norm of an algebraic integer.

Lemma 4.1 (An inert prime in an interval)**.** *Suppose that $D > 331$, $h(-D) = 1$, and $D < A < B$. For $y \ge 1$, put $c_y = 1$ when y is odd and $c_y = 1/2$ when y is even. Define*

$$E_D(A,B) = \sum_{y=1}^{\lfloor\sqrt{4B/D}\rfloor} \left[1 + c_y\left(\sqrt{B - Dy^2/4} - \sqrt{\max(0, A - Dy^2/4)}\right)\right]. \tag{4.1}$$

If $E_D(A,B) < \pi(B) - \pi(A)$, then $(A,B]$ contains an inert prime.

Proof. The elementary reduction makes D prime, so every prime in $(A,B]$ is unramified. If such a prime p splits, a principal ideal of norm p gives an element $x + y\omega$ with

$$p = x^2 + xy + \frac{D+1}{4}y^2 = \left(x + \frac{y}{2}\right)^2 + \frac{D}{4}y^2.$$

Here $y \ne 0$, since a rational integer has square norm. Changing the sign and taking the conjugate if necessary, we may assume $y > 0$ and $x + y/2 \ge 0$. Therefore $y \le \sqrt{4B/D}$, and, for a fixed y, $x + y/2$ lies between the two square roots in (4.1). An interval of length L contains at most $1 + L$ members of any translate of $\mathbb{Z}$.

When y is even, we can improve this count. Writing $y = 2k$ gives

$$x^2 + xy + \frac{D+1}{4}y^2 \equiv x^2 \pmod 2.$$

Since p is odd, x is odd. Its permitted values are consequently spaced by two, and an interval of length L contains at most $1 + L/2$ of them. Summing these bounds gives at most $E_D(A,B)$ split primes. If this is smaller than the number of primes in the interval, one of them must be inert. □

4.1. **The three parameter choices.** The six discriminants require three choices of polynomial space and node count, given in Table 1. Throughout we set

$$S = \left\lfloor \frac{6P}{5K} \right\rfloor, \qquad R = KS, \qquad J = \lfloor \varepsilon_0 R \rfloor. \tag{4.2}$$

Here P is the prime cutoff in the arithmetic estimate, and ε_0 is a rational number; the actual ratio in the size estimate is $\varepsilon = J/R$.

D	P	K	ε_0	n
$419, 443, 467, 491$	$312\,000\,000$	43	$1/8$	$35\,155\,200$
379	$312\,000\,000$	17	$16/125$	$86\,265\,600$
347	$756\,000\,000$	17	$16/125$	$208\,919\,200$

Table 1. Parameters for the six determinant comparisons. Each row specifies S, R, J through (4.2).

The ratios $\theta = M/(nJ)$ in the three rows are approximately 0.990690, 0.997262, and 0.997782. Thus these choices use only about 0.94%, 0.27%, and 0.22% more nodes than the necessary minimum M/J. The exact rank criterion is needed at these counts: the coarser sufficient inequality in [2, Theorem 5.1] does not hold uniformly over the corresponding auxiliary-prime windows.

Table 2 gives the intervals used to supply an inert prime. For the first row we use the weaker count obtained by setting all $c_y = 1$; it already suffices. Its displayed split-prime bound is the largest of the four separately evaluated counts, not a count on the whole interval $419 \le D \le 491$.

D	$(A, B]$	$\pi(B) - \pi(A)$	Split-prime bound	$2\rho_n <$
419, 443, 467, 491	(200 000, 228 000]	2276	2181.456	22 281
379	(500 000, 515 000]	1135	990.602	32 708
347	(800 000, 820 000]	1465	1366.923	49 785

TABLE 2. Upper bounds for the number of split primes and the node diameter. All decimal bounds are rounded upward.

In each row $D < A$, and the split-prime bound is smaller than the prime count. Lemma 4.1 therefore supplies an inert prime $p \in (A, B]$. The diameter bound shows that the nodes remain distinct modulo $p\mathcal{O}_D$: a nonzero element of that ideal has modulus at least p, whereas a difference of two nodes has modulus less than A. Moreover $K < A < B < R < A^2$, so $K < p < R < p^2$.

The remaining conditions in Theorem 3.1 are the finite integer inequalities $n\mu_e(p) > B_e(p)$. They hold for every prime in the corresponding window and every $0 \le e \le \lfloor (R-1)/p \rfloor$. The least differences $n\mu_e(p) - B_e(p)$ for the three rows are respectively

$$13\,425\,248\,140, \qquad 6\,204\,049\,940, \qquad 89\,748\,485\,986.$$

The sums defining these integers are evaluated directly; their complete ranges and a short integer-arithmetic program are in the repository. Thus the particular inert prime supplied by the count satisfies every hypothesis of the rank theorem, and V has full row rank in all six cases.

5. Completion of the proof

It remains to evaluate the determinant comparison at the parameters of Table 1. For the size estimate we use

$$(a, v, \beta, a_0) = \left(\frac{39}{40}, \frac{19}{50}, -1, \frac{1}{200}\right), \qquad (N_t, N_z, N_m) = (128, 256, 512). \tag{5.1}$$

The last two integers are the mesh sizes for the column-order capacity bound and the intervals of possible m/R. The finite choices of affine composition parameters are the same as in Appendix C of [2]. These values satisfy the finite hypotheses of Section 2. At each upper radial endpoint, the calculation checks the moment and concavity conditions before using the Schur bound.

For the arithmetic estimate we retain only the two- and three-factor composite corrections. The cutoff for the derivative slope is

$$q_{\mathrm{cap}} = \min\{J - 1,\, m_D^4 - 1,\, m_D \lfloor P/m_D \rfloor\}, \qquad m_D = (D+1)/4.$$

It is 46 799 995 in the first parameter family, 47 923 198 at $D = 379$, and 57 289 760 at $D = 347$. In each case the factor bounds and column capacities in the arithmetic estimate are satisfied.

Table 3 gives rigorous enclosures for the difference $G = \mathcal{A} - \mathcal{U}$ between the arithmetic lower bound and the size upper bound in Proposition 2.1. The calculations use directed rounding, and the displayed endpoints are rounded outward.

D	Enclosure for G
347	[0.00146083, 0.00147313]
379	[0.01186524, 0.01187031]
419	[0.00417435, 0.00417636]
443	[0.04790491, 0.04790692]
467	[0.11463345, 0.11463546]
491	[0.17810049, 0.17810250]

TABLE 3. The six positive comparison margins, with two- and three-factor composite corrections.

Proof of Theorem 1.1. Suppose $h(-D) = 1$ and $D > 331$. The bound $D \leq 498$ from [2] and the elementary reduction in Section 1 restrict D to $347, 379, 419, 443, 467, 491$. Section 4 supplies the auxiliary inert prime and verifies the hypotheses of Theorem 3.1, so the interpolation matrix has a nonzero maximal minor in each case. Proposition 2.1 requires $G \leq 0$, whereas every lower endpoint in Table 3 is positive. This excludes all six possibilities. □

The bound 331 shows that a sharper nonvanishing criterion allows the existing determinant estimates to go further. The arithmetic estimate still uses class number one to control non-inert primes by norm representations. A general argument for Landau–Siegel zeros would require a substitute for that input.

AI USE

The underlying research was carried out by autonomous rStation agents using Gemini 4 Argon and GPT-6 Astra. The author launched the agents, which conducted the research without human intervention. The sources for the earlier bound are paper 2, version 12 and paper 22, version 1. The source for the bound 331 is rStation paper 24. Its calculations extend paper 2, version 13, together with the rank argument developed in paper 22, version 4. The spectral estimates originate in paper 16, version 2.

Separate agents using GPT-6 Astra and Claude Opus 5.5 checked the original arguments independently of the discovering agents. GPT-6 Astra, Claude Opus 5.5 and Gemini 4 Argon were then used in preparing the present exposition, including mathematical consultation, revision of the outline and prose, and review of the proof. GPT-6 Astra recomputed the six numerical comparisons for this paper and used separate programs to check the rank inequalities, the auxiliary-prime counts and the remaining discriminants. The source calculation at 347 included a four-factor composite correction; the comparisons here use only the two- and three-factor corrections.

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